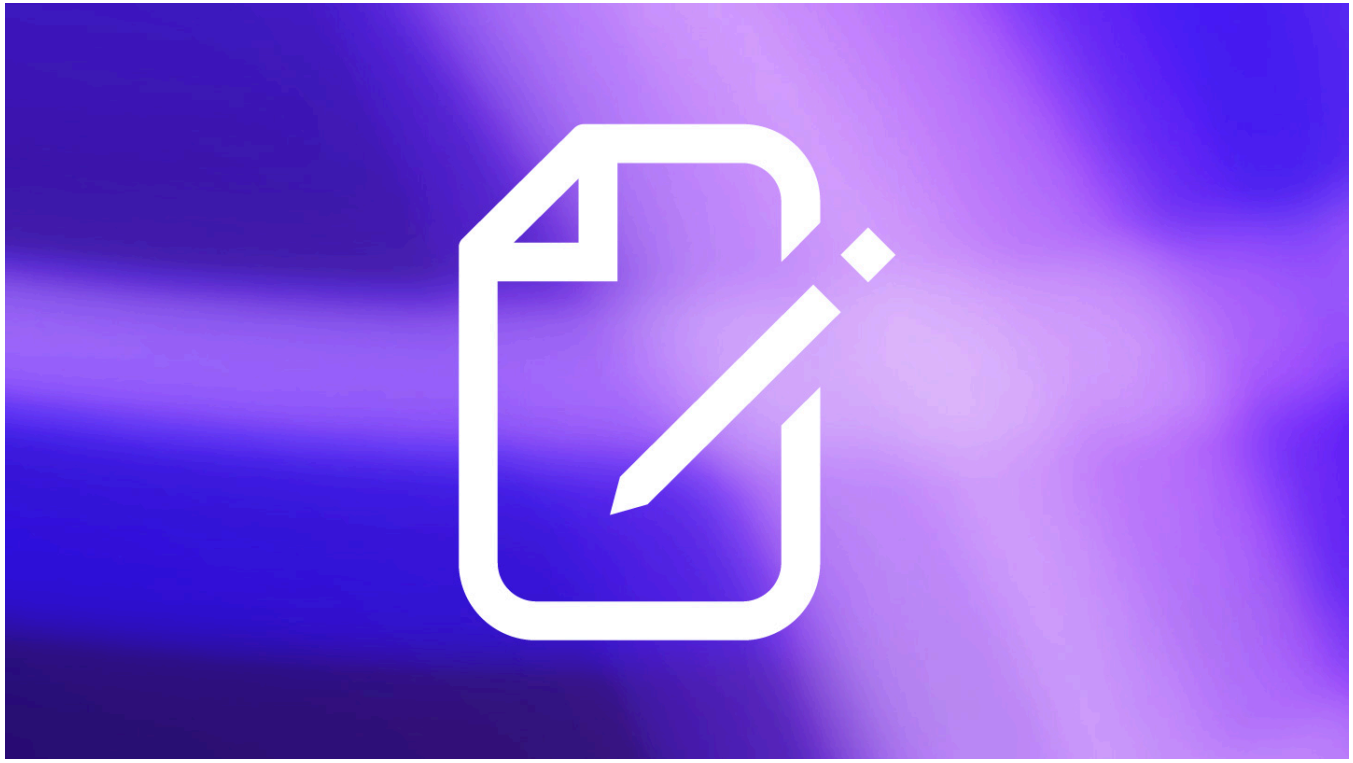
National

Operational Excellence Standard (OES) — Connected Devices Insights

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Summary

The Connected Devices Insights feature in Tech360 enables the technician to view all devices connected to the customer's gateway, including interactive filters and in-depth telemetry to help prioritize triaging potentially impaired devices.

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What's changed in this article

Date	Summary of change
December 10, 2025	Added to Celestial.
July 1, 2022	Conducted annual review.
June 1, 2021	Released document.

Overview

Connected Devices Insights provides the technician a snapshot of all the devices on the customer's WiFi network provided by a Comcast device, as well as their wired devices. Customer-owned WiFi routers, Access Points (APs), or mesh devices may not be displayed.

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Connected Devices Insights — Telemetry features

The in-depth telemetry insights that are included in the feature are:

- PHY rates
- Received Signal Strength Indicator (RSSI)
- Gateway information
- Speed to gateway
- Service Set Identifier (SSID)
- Signal to Noise ratio (SNR)
- Wireless device information
- Wired device information

Telemetry definitions

PHY rates

- PHY refers to a physical communication between two or more IP network devices. PHY Rate refers to the speed the devices are communicating measured in Mbps.
- In a standard home network, there is usually a gateway responsible for managing multiple PHY Rate sessions known as “streams."
- If you look at the history on a device and see low PHY rates here or there; this is normal. The device may be in standby, asleep or just not transferring data at those times.

RSSI

- RSSI stands for Received Signal Strength Indicator.

- RSSI is a WiFi signal strength indicator.
- While RSSI is a good measure of signal strength, it's important to remember it doesn't necessarily reference quality.
- Strength and quality of the signal are just as important for WiFi. Use RSSI as a leading indicator of strength only

SNR

- SNR refers to the Signal to Noise ratio or the strength of a signal relative to the background noise or interference.
- A higher SNR indicates a stronger, clearer signal, and signifies less noise interference and better signal quality

Gateway device information

- Make and model of the gateway.
- Current firmware.

SSID

- SSID refers to the Service Set Identifier or the broadcasting name of the wireless network used to connect wirelessly to the gateway network.

Speed to gateway

- Speed to gateway measures the speed of the internet connection between the upstream network (head end) and the customer's gateway. Customer-owned and managed (COAM) is not supported.
- This test determines speeds before going through the WiFi conditions in a customer's premise home or through various devices.
- A test failure is defined as customers who runs two failing speed tests in the Xfinity app — receiving less than 90% of advertised speed after a restart.
- The results of the speed to gateway test could be impacted by variables, such as:
  - In premise or drop issues.

- Network issues.
- Gateway issues.

Wireless devices

- Wireless devices refers to any client device that is connected to the gateway via IEEE 802.11 standards.

Wired devices

- Wired devices refers to any client device that is connected to the gateway ethernet network via IEEE 802.3 standards.

Technician guidance

Access the Connected Devices Insight Tool

The tool is available through Tech360 in the **Account Health** tab, under **Connected devices**.

When to use the Connected Devices Insight Tool

The Connected Devices Insight Tool should be used on all jobs where a customer is reporting slow speeds and/or intermittent connectivity issues, that are not related to modem/gateway performance based on RF signal quality issues.

Connected devices supplemental workflow

1. **Troubleshooting:**
   - Technician performs normal troubleshooting processes to identify if RF impairments exist between the tap and gateway and validate that all OES requirements are met.

- Use the OES Refer to Maintenance (RTM) process for any signal impairments identified at the tap.
- During the normal troubleshooting processes, the technician shall utilize their XM Meter to verify correct speeds at the tap, bond block, and outlet.
- If no RF or speed impairments exist, the technician will run an XM Meter speed test at the gateway outlet via Ethernet to verify device health.

2. **Perform connected devices test:**
   - Once test completes, review devices identified as impaired.
   - Determine which impaired device(s) the customer is experiencing issues with.
   - Review the reported errors, if multiple devices are showing errors compare errors and see what may be in common.
   - Review the "*Check for the following*" suggestions and follow the recommended actions. Perform common actions first if multiple devices are affected.
   - Review the actions with the customer before making any changes.
   - If the customer declines the suggested actions review with the customer how performance could be affected and why.
   - Once recommended actions have been performed, refresh the connected devices test and review the results.
   - Inform the customer of the results and any further actions that may be needed.
   - Resume normal troubleshooting and work flow processes.

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